

Verification and Integrity Specifications

This document specifies the deep integrity verifications and ontology checks executed by the RAG pipeline's checking commands.

Deep Integrity Verification: `rag verify`

Symmetrically, `run_verify` is a mandatory, sequential checker. It executes a comprehensive, deterministic audit over the entire processed catalog and filesystem on disk to certify that all files, hashes, crops, and database records match with absolute precision. Symmetrically, if any checks fail, it blocks the catalog export step.

A. Ontology Tamper Check (Gate 1 Freeze)

- **Mechanism:** Re-computes the SHA-256 hash of the active layer ontology file inside the library folder.
- **Validation:** Compares this hash against the `layers_hash` metadata field stored inside the document inventory database.
- **Failure Behavior:** If the ontology file was deleted or its contents changed since document classification, the validator reports ontology tampered—enforcing the invariant that the ontology remains immutable once the pipeline starts.

B. Corpus File Hash Verification

- **Mechanism:** Re-calculates the actual SHA-256 hash of every PDF inside the corpus.
- **Validation:** Matches the current hash against the \$12\$-character short hash recorded in the document inventory database.
- **Failure Behavior:** If a PDF was modified or swapped on disk, it outputs `HASH CHANGED` and fails verification.

C. Crop Files Existence Validation

- **Mechanism:** Iterates over every asset listed inside the asset inventory database that carries a high-resolution crop reference.
- **Validation:** Verifies that the corresponding crop PNG physically exists on disk under the promoted crops directory.

- **Failure Behavior:** Missing crop files are reported as failures, halting the export validation.

D. Orphan Crop Detection

- **Mechanism:** Scans the promoted crops directory and matches every PNG file against the crops claimed inside the regions database.
- **Validation:** Identifies any "orphan" crops that are no longer referenced by any region in the manifest.
- **Failure Behavior:** Flags them as warnings. Orphans represent leftover clutter and do not block final catalog exports.

E. Region Links and Approval Validation

- **Mechanism:** Scans every asset listed inside the asset inventory database carrying a region ID.
- **Validation:** Verifies that the corresponding coordinate region exists inside the regions database and holds human approval.
- **Failure Behavior:** Flags missing or unapproved region links as failures.

F. Completion Metrics Summary

- **Mechanism:** Evaluates and displays the current counts for computer-verified and human-approved assets.
- **Result:** If all checkers pass, it touches the verification success marker file inside the logs directory and prints a success statement declaring the assets are approved and ready for export.